MILITARY SPECIFICATION SHEET

CONNECTOR, ELECTRICAL, CIRCULAR, MINIATURE, COMPOSITE, HIGH DENSITY, QUICK COUPLING, ENVIRONMENT RESISTANT, REMOVABLE CRIMP CONTACTS, BACKSHELL, SHIELD TERMINATING, (SERIES A AND B)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the backshells described herein shall consist of this specification and the latest revision of MIL-C-29600.

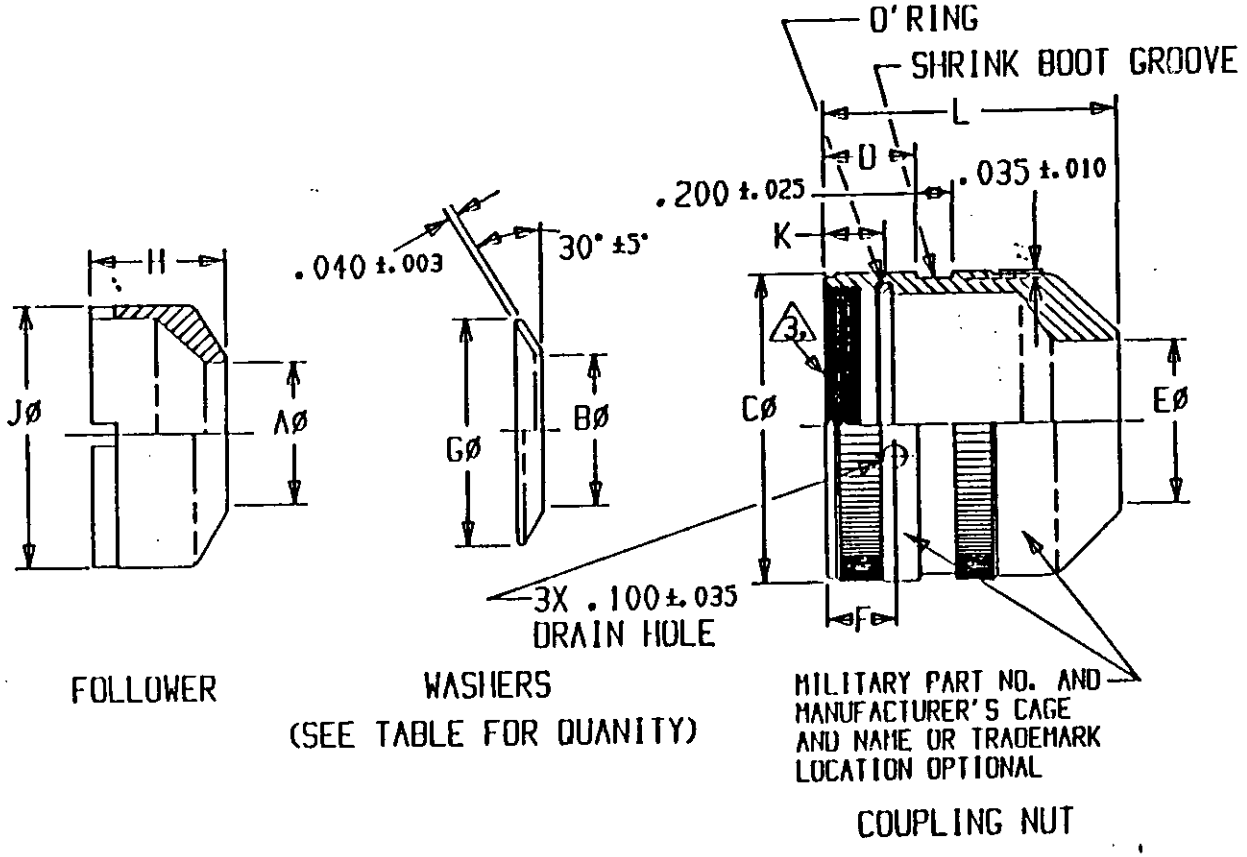


FIGURE 1. Backshell, Straight, Shield terminating, Environmental.

NOTES:

1. Dimensions are in inches unless otherwise specified.
2. Metric equivalents are based upon 1 inch = 25.4 mm.
3. See MIL-C-29600 for interface dimensions.

AMSC N/A FSC 5935

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Shell Size	G ±.010	H Ref.	JØ ±.009
08/09	.354	.579	.406
10/11	.479	.593	.531
13	.606	.590	.656
14/15	.729	.620	.781
16/17	.854	.638	.906
18/19	.979	.642	1.031
20/21	.979	.670	1.156
22/23	1.219	.687	1.281
24/25	1.219	.692	1.406

TABLE I. Dimensions.

Connector Type	Shell Size	A ±.010 Over Wire Bundle	B ±.010 Over Gross Shield	Number of Washers	C Max.	D ±.010	E ±.010 Over Cable Jacket	L ±.015	K Max	MS9068 dash No.
MIL-C-29600 Series B	8	.156	.203	3	.661	.400	.265	1.193	.280	-013
	10	.234	.265		.789		.296			-015
	14	.390	.437		1.044		.500			-019
	16	.453	.515		1.171		.562			-021
	18	.562	.640		1.282	.410	.671			-023
	20	.593	.640		1.410	.475	.703	1.513	.318	-025
	22	.656	.812		1.537		.750			-027
	24	.765	.812		1.668		.875			see fig.2
MIL-C-29600 Series A	9	.156	.203	4	.661	.400	.265	1.193	.218	-013
	11	.234	.265		.789		.296			-015
	13	.387	.415		.916		.451			-017
	15	.390	.437		1.044		.500			-019
	17	.453	.515		1.171		.562			-021
	19	.562	.640		1.282	.410	.671			-023
	21	.593	.640		1.410	.475	.703	1.513	.254	-025
	23	.656	.812		1.537		.750			-027
	25	.765	.812		1.668		.875			see fig.2

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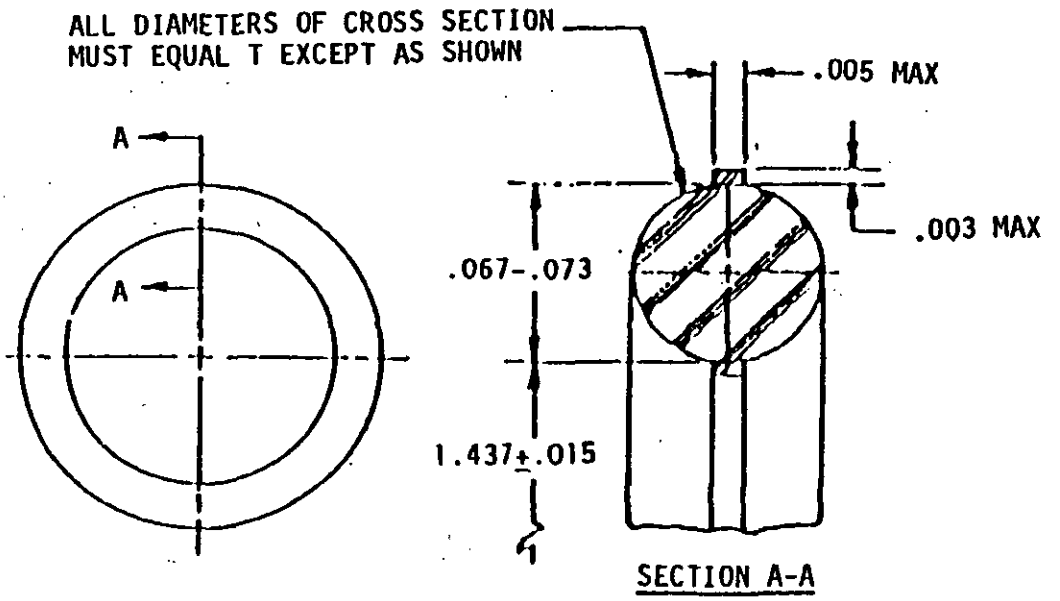


FIGURE 2

Inches	mm	Inches	mm	Inches	mm
.003	0.076	.437	11.10	.729	18.52
009	0.229	.451	11.46	.750	19.05
.010	0.254	.453	11.51	.765	19.43
.012	0.305	.475	12.07	.781	19.84
.015	0.381	.479	12.17	.789	20.04
.025	0.635	.500	12.70	.812	20.62
.035	0.889	.515	13.08	.854	21.69
.040	1.016	.531	13.49	.875	22.23
.156	3.962	.562	14.27	.906	23.01
.200	5.080	.579	14.71	.916	23.27
.203	5.156	.590	14.99	.979	24.87
.218	5.537	.593	15.06	1.031	26.19
.234	5.944	.606	15.39	1.044	26.52
.254	6.452	.620	15.75	1.156	29.36
.265	6.731	.638	16.21	1.171	29.74
.280	7.112	.640	16.26	1.193	30.30
.296	7.518	.642	16.31	1.219	30.96
.318	8.077	.656	16.66	1.281	32.54
.354	8.992	.661	16.79	1.282	32.56
.387	9.830	.670	17.02	1.406	35.71
.390	9.906	.671	17.04	1.410	35.81
.400	10.16	.687	17.45	1.513	38.43
.406	10.31	.692	17.58	1.537	39.04
.410	10.41	.703	17.86	1.668	42.37
.415	10.54				

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TABLE II. Qualification tests. 1/

Inspection	Requirement Paragraph	Test Paragraph
Examination of product	3.1 thru 3.5, 3.40 and 3.41	4.6.1
EMI shielding effectiveness	3.27	4.6.22
Temperature cycling	3.8	4.6.3
Vibration	3.22	4.6.17
Shock	3.23	4.6.18
Shield to shell conductivity	MIL-C-29600/2	4.6.19.3
External bending moment	3.17	4.6.12
Salt spray (2000 hours)	3.15	4.6.10.1
Accessory thread strength	3.39	4.6.35
Impact	3.38	4.6.33
Fluid immersion	3.29	4.6.24
Coupling torque	3.9	4.6.4
Plating adhesion	3.28	4.6.32
Examination of product	3.1 thru 3.5, 3.40 and 3.41	4.6.1

1/ Unless otherwise specified, all qualification tests shall be performed in accordance with MIL-C-29600.

REQUIREMENTS:

Requirements shall conform to MIL-C-29600.

Dimensions: See Table I.

Design characteristics and configuration: See Figure 1.

O-ring: Use MS9068 per table.

Material: Material shall be as specified in MIL-C-29600, paragraph 3.3.2.

Finish: Finish shall be as specified in MIL-C-29600, paragraph 3.3.7.

Manufacturer's identification: All components of this backshell shall contain the manufacturer's CAGE No. and name or trademark. Marking of washers and followers is not required. All components of the final backshell assembly shall be of the same manufacturer.

Shield to shell conductivity: When tested as specified in MIL-C-29600, paragraph 4.6.19.3, the maximum resistance measured between the shield terminated to the connector plug and the receptacle flange shall be 20 milliohms.

Qualification: See Table II.

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REQUIREMENTS (continued):

Shell size ranges: small 8-13, medium 14-19, large 20-25.

Sample size: Two samples in each shell size range and each finish.

Sample preparation: Each sample shall be wired to applicable wired connectors or equivalent hardware.

Application: For use with MIL-C-29600 Series A and B connectors.

Example of part number:



Custodians:
Army - CR
Navy - AS
Air Force - 85

Preparing activity:
Navy - AS
(Project No. 5935-3701-14)

Review activities:
Army - AV
Navy - EC
Air Force - 15
DLA - ES
NASA - NA

Users:
Army - MI
Navy - SH,MC
Air Force - 15

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